



S2.2 The covalent model

Structure 2 · Models of bonding and structure · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A covalent bond is a shared pair of electrons between two atoms, formed to achieve a more stable (often noble-gas-like) electron arrangement; Lewis (electron dot) structures show all bonding and non-bonding valence electron pairs, and can be used with VSEPR theory (electron domains repel to be as far apart as possible) to predict molecular shape and bond angles. Bond polarity arises from differences in electronegativity between bonded atoms, and the overall molecular polarity depends on both bond polarities and molecular shape (symmetric arrangements can cancel out individual bond dipoles). Covalent molecules are held together internally by strong covalent bonds but interact with neighbouring molecules only through weaker intermolecular forces (London/van der Waals forces, dipole-dipole forces, hydrogen bonding), which determine bulk properties such as boiling point.

Worked example: follow the reasoning

Deduce the shape and bond angle of an ammonia molecule, NH_3 .

Step 1: N has 5 valence electrons; each of 3 H atoms contributes 1 electron, giving 3 N-H bonding pairs and 1 lone pair on N (total 4 electron domains).

Step 2: VSEPR: 4 electron domains arrange tetrahedrally to minimise repulsion, but the lone pair repels bonding pairs more strongly than bonding pairs repel each other.

Step 3: This compresses the H-N-H angle slightly below the ideal tetrahedral angle.

Answer: NH_3 has a trigonal pyramidal shape with a bond angle of approximately 107° (less than the tetrahedral 109.5° due to lone pair-bond pair repulsion).

Quick check · multiple choice

Which molecule is non-polar overall despite containing polar bonds?

- A H_2O
- B NH_3
- C CO_2
- D CHCl_3

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider the molecule boron trifluoride, BF_3 , and the ion tetrafluoroborate, BF_4^- . [8]
 - (a) Draw the Lewis structure of BF_3 and state its molecular shape and F-B-F bond angle. [3]
 - (b) Explain why BF_3 acts as a good electron-pair acceptor (Lewis acid). [2]
 - (c) Deduce the shape of BF_4^- formed when BF_3 accepts a fluoride ion, F^- , and explain the change in bond angle compared with BF_3 . [3]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) B is bonded to 3 F atoms, each with 3 lone pairs, and B has no lone pair (only 6 electrons/3 bonding pairs around B). Shape: trigonal planar. Bond angle: 120° . [3: correct structure, shape named, angle stated]

(b) B has only 6 electrons in its outer shell (an incomplete octet) and an empty orbital, so it can accept a lone pair of electrons from a species such as F^- to complete its octet, acting as a Lewis acid (electron-pair acceptor). [2]

(c) BF_4^- has 4 bonding pairs and no lone pairs around B, so the electron domains arrange tetrahedrally to minimise repulsion; shape is tetrahedral with bond angle 109.5° , which is larger than in BF_3 because there are now 4 equivalent bonding-pair repulsions (all equal) arranged tetrahedrally instead of 3 bonding pairs arranged trigonal planar. [3]

Quick check answer

C. CO_2 is linear and symmetric, so the two equal and opposite $C=O$ bond dipoles cancel exactly, giving no overall molecular dipole, despite each $C=O$ bond being polar.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.